Pakistan IDF Atlas

PRECIPITATION FREQUENCY ANALYSIS

TECHNICAL REPORT

Observed precipitation and stationary IDF assessment

SYNTHETIC DEMONSTRATION — not measured rainfall · uploaded gauge series · district independent

Stationarity not rejected at α = 0.05

REPORT IDENTIFIER **PK-IDF-OBS-SYNTHETIC-DEMONSTRATION-NOT-MEASURED-RAINFALL-STA**	ISSUE DATE **11 November 2026**
PREPARED FOR **Hydrologists, planners and engineering reviewers**	REVISION **2.0 · generated from active analysis**
PRECIPITATION SOURCE **SYNTHETIC DEMONSTRATION — not measured rainfall · observed daily**	ADMINISTRATIVE BOUNDARY **Not used for uploaded gauge fitting**
SELECTED MODEL **Stationary · parameters constant**	SELECTED METHOD **GEV · L-moments**

PROFESSIONAL TECHNICAL REPORT

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1.0 Executive summary

The uploaded daily record produced 35 eligible 24-hour annual maxima. Using GEV · L-moments, the estimated 24-hour intensity is 3.18 mm/hour for the 10-year event and 4.54 mm/hour for the 500-year event. The 250- and 500-year results extend far beyond the record and require independent review.

Assessment at a glance

Result status

Stationarity not rejected at α = 0.05

Basis of assessment

GEV · L-moments · stationary parameters. 35 eligible 24-hour annual maxima.

Duration support

Daily data support 24-hour totals. Any shorter duration is derived and must be checked against local evidence.

Required next action

Check source quality, duration assumptions and model sensitivity; obtain independent hydrologic review before design adoption.

Decision summary · selected 24-hour return levels

RETURN PERIOD (YEARS)	ANNUAL EXCEEDANCE PROBABILITY	INTENSITY (MM/HOUR)	24-HOUR DEPTH (MM)
10	10.0%	3.18	76.32
50	2.0%	3.81	91.44
100	1.0%	4.04	96.96
250	0.4%	4.33	103.92
500	0.2%	4.54	108.96

10-YEAR · 24-HOUR
3.18 mm/hr
Estimated intensity

50-YEAR · 24-HOUR
3.81 mm/hr
GEV · L-moments

100-YEAR · 24-HOUR
4.04 mm/hr
Stationary model

250-YEAR · 24-HOUR
4.33 mm/hr
Rare-event extrapolation

500-YEAR · 24-HOUR
4.54 mm/hr
Highest uncertainty · 0.2% AEP

2.0 Purpose and how to read this report

This report estimates rainfall intensity for selected storm durations and annual exceedance probabilities from the active dataset. It documents the data used, the checks applied, the fitted method, the uncertainty and the limits on engineering use. It is a technical analysis record, not regulatory approval or independent design certification.

What was analysed?

The uploaded daily gauge series for SYNTHETIC DEMONSTRATION — not measured rainfall. District boundaries are not used in the calculation.

What does “T-year” mean?

A T-year value has a 1/T chance of being exceeded in any year. For example, a 100-year value has 1% annual exceedance probability; it does not occur on a fixed 100-year schedule.

Which durations are measured?

Daily data directly support 24 hours only. All shorter durations are derived with the stated scaling equation.

How should rare values be used?

Results for 250 and 500 years are model extrapolations well beyond most records. Use them only with uncertainty, method sensitivity and independent hydrologic review.

3.0 Data sources and quality

Spatial treatment: The uploaded gauge series is fitted directly. District boundaries do not enter quality control, annual-maxima extraction, distribution fitting, uncertainty, or IDF calculations.

3.1 Source provenance

All compatible worksheets in the user-selected Excel or CSV file are scanned and processed locally. The report retains the source label, native interval, date range, quality-control counts, optional gauge coordinates and precipitation unit. District selection and polygon averaging are disabled, and the workbook is not transmitted to a server.

Table 3-1. Analysis population, provenance and metric definitions

PRECIPITATION SOURCE	SYNTHETIC DEMONSTRATION — not measured rainfall · observed daily	User-supplied observed daily precipitation
TEMPORAL SUPPORT	Daily totals	Native support: 24-hour duration
ANALYSIS PERIOD	1985-01-01 to 2019-12-31	Selected range applied before QC and fitting
SPATIAL ASSIGNMENT	Uploaded station / gauge series	District selection disabled; no boundary used in fitting
ANALYSIS SAMPLE	12,783 accepted daily rows; 35 eligible annual maxima	100.00% span completeness; 0 missing daily intervals; 0 duplicates; 0 invalid values
RESPONSE METRIC	Rainfall intensity (mm/hour)	Precipitation depth divided by duration in hours
RETURN PERIODS	2, 5, 10, 25, 50, 100, 250 and 500 years	AEP=1/T; 250- and 500-year levels are high-uncertainty extrapolations

3.2 Quality-control rules

Dates are sorted and de-duplicated, non-numeric or negative precipitation values are rejected, missing calendar days are counted, and only years with at least 90% expected daily coverage are eligible for the annual-maxima series. These automated checks support traceability but do not replace metadata-led homogenization of station relocations, instruments or observation practices, consistent with WMO climate-data guidance.

4.0 Methodology

The workflow follows an annual-maxima frequency-analysis structure commonly used in hydrology (Hosking, 1990; Katz et al., 2002). Equation definitions are reproduced in Appendix A.

01
Ingest and screen
Dates, units, range and completeness

02
Build AMS
Eligible one-day annual maxima

03
Fit and diagnose
Distributions, trend and uncertainty

04
Convert to IDF
Depth-duration scaling and intensity

4.1 Literature and implementation alignment

Table 4-1. Literature basis, implementation status and interpretation

ANALYSIS COMPONENT	LITERATURE OR GUIDANCE	IMPLEMENTED USE	TECHNICAL BOUNDARY
DATA QUALITY AND HOMOGENIZATION	WMO climate-data guidance	Automated range, duplicate, missing-day and annual-completeness checks	Station history and neighbour-based homogenization still require expert review
ANNUAL MAXIMA AND EXTREMES	Katz et al. (2002)	Calendar-year one-day annual-maxima series and return-level estimation	Block-maxima results depend on record length and independence
STATIONARY DISTRIBUTION SUITE	Hosking (1990); Katz et al. (2002); USGS Bulletin 17C for LP3 context	Six alternatives fitted to the same annual-maxima sample: GEV by L-moments; Gumbel, Lognormal, Gamma and Weibull by moment-based estimates; LP3 in log space	The lowest RMSE is a screening preference only; probability plots, uncertainty and tail sensitivity remain essential
NONSTATIONARY GEV	Cheng and AghaKouchak (2014); Agilan and Umamahesh (2017, 2018); Kim et al. (2017)	Maximum-likelihood GEV with $\mu(y)=\mu_0+\mu_1(y-\bar{y})$, constant σ and ξ; stationary and nonstationary MLE compared by AICc	Time is a statistical index, not automatically a physical cause; covariate and parameter uncertainty remain material
RARE RETURN PERIODS	Wehner et al. (2024); Patel and Schneider (2026)	250- and 500-year estimates, bootstrap intervals and explicit extrapolation warnings	Short records can strongly bias ξ and rare return levels; intervals exclude model-form uncertainty
IDF DURATION RELATIONSHIP	Koutsoyiannis et al. (1998)	Transparent power-law duration scaling followed by conversion to mm/hour	Daily data directly support 24 hours only; shorter durations are derived
TREND ASSESSMENT	Mann (1945)	Two-sided Mann–Kendall test retained as a diagnostic, not as the nonstationary model	No serial-correlation correction; non-rejection is not proof of stationarity
PAKISTAN CLIMATE DRIVERS	Masood et al. (2023); Hussain et al. (2023)	Annual ENSO and other candidate drivers can be uploaded, physically documented, aligned to eligible maxima and compared with time and stationarity	Relationships vary by precipitation index, season, elevation and station; no universal coefficient is assumed and correlation alone cannot select a covariate
LP3 COMPARISON	USGS Bulletin 17C	Log-Pearson III benchmark in the stationary multi-method table	Not full Bulletin 17C compliance or a flood-frequency implementation

4.2 Infrastructure design-return-period guide

Table 4-2. Literature-reported return periods and typical infrastructure applications

RETURN PERIOD	AEP	TYPICAL INFRASTRUCTURE	HOW TO USE IT	LITERATURE BASIS
2–5 YEARS	50%–20%	Minor local drainage, small roadside drains and nuisance-flood conveyance	Frequent-service screening; verify municipal criteria and allowable surface flooding	International road-drainage practice; project authority governs
10 YEARS	10%	Rural road drains and minor culverts	Common minimum for minor rural crossings	Low Volume Rural Road Manual, Part C (2019)
25 YEARS	4%	Urban road drainage and culvert hydraulic design	Pakistan project example uses 25-year design for culverts	AIIB Rawalpindi–Hassanabdal assessment (2025)
50 YEARS	2%	Culvert capacity and resilience check	Check that the structure passes the larger event without unacceptable damage	AIIB Rawalpindi–Hassanabdal assessment (2025)
100 YEARS	1%	Bridges, major crossings, floodplain development and low-hazard dam screening	Pakistan project example uses 100 years for bridges; consequence assessment remains essential	AIIB (2025); ICOLD Bulletin 156
250–500 YEARS	0.4%–0.2%	High-consequence urban or industrial flood protection and critical-infrastructure sensitivity checks	Use as an upper-tail stress test only where the owner adopts a risk-based criterion; not a universal design requirement	ICOLD Bulletin 156
BEYOND 500 YEARS / PMF	<0.2% / not assigned	High- and extreme-hazard dams	The atlas stops at 500 years and is not sufficient for high-hazard dam design	World Bank Good Practice Note on Dam Safety (2021)

> **Design-authority warning.** This table is a literature-based orientation aid, not a Pakistan-wide code prescription. The governing authority may require a different event, a separate check flood, climate-change allowance, freeboard, blockage/scour analysis, or a consequence-based standard. Rainfall return period is also not automatically identical to the resulting flow or flood return period.

4.3 Stationary, nonstationary and reference-year interpretation

Stationary means the probability distribution used for annual precipitation maxima has parameters that do not change with time or another predictor. A 100-year intensity therefore has the same model value in every year. **Nonstationary** means at least one parameter or return level changes with a covariate such as time, temperature, ENSO or urbanization. The return level is conditional on a stated covariate value, so it must be reported with that value rather than as a timeless quantity.

Table 4-3. How to calculate, select and interpret the evaluation point

SELECTION	CALCULATED VALUE / USE	INTERPRETATION	WARNING
AUTOMATIC RECORD-CENTRE	$\bar{y}$ = Σ eligible AMS years / n = 70070 / 35 = 2002.00	Default evaluation point; the arithmetic mean of eligible AMS years makes the centered-year covariate zero	It is a statistical reference time, not automatically the asset design year

SELECTION	CALCULATED VALUE / USE	INTERPRETATION	WARNING
LATEST ELIGIBLE AMS YEAR	2019	Evaluates conditions at the latest year supported by an eligible annual maximum	More sensitive to slope uncertainty than the centre
CUSTOM YEAR INSIDE AMS SPAN	1985–2019	Interpolation of the implemented time relationship	Still conditional on the assumed linear model
CUSTOM YEAR OUTSIDE AMS SPAN	Use only for a stated planning horizon	Extrapolation beyond the data	Not a climate projection; uncertainty grows rapidly
CMIP6 PERIOD MIDPOINT	2055	Label for the selected 20-year time slice	Not a single-year observation or a fitted reference year
MATCHED COVARIATE EVALUATION	Requires a retrieved or uploaded annual covariate series	Conditional return level at the selected mean, latest or custom annual covariate value	The coefficient is fitted from matched annual data and reported with a 95% moving-block bootstrap interval

Reference-year rule. The software default is calculated automatically as $\bar{y}=\Sigma y_i/n$ using only eligible annual-maxima years. Centering is a regression re-parameterization: it makes μ_0 the fitted location at the centre of the observed time covariate without changing μ_1 or the fitted trend. Cheng and AghaKouchak (2014) and Sarhadi and Soulis (2017) formulate GEV parameters as functions of time; Agilan and Umamahesh (2017, 2018) emphasize covariate choice and uncertainty. Literature does not prescribe one universal future reference year. A latest or custom year is therefore an evaluation choice, and a year outside the eligible AMS span is explicitly classified as extrapolation rather than a climate projection.

4.3 Calculation sequence and model definitions

Ingestion and QC. Parse the selected native timestamp and precipitation columns; apply the custom date range; retain one record per day or hour; reject non-numeric or negative precipitation; and inventory missing, duplicate and invalid records.

Annual-maxima series. Retain each calendar year with at least 90% coverage at the selected native interval. Daily input yields one-day maxima; hourly input yields rolling maxima for each supported 1- to 24-hour duration. The software requires at least 8 eligible maxima for fitting and at least 30 before issuing a diagnostic preference based on the lowest RMSE.

Stationarity assessment. Apply the two-sided Mann–Kendall monotonic-trend test at $\alpha=0.05$ and estimate Sen's median slope (Mann, 1945). Lag-1 correlation is reported as an independence diagnostic; the p-value is not adjusted for serial correlation.

Stationary probability distributions. Fit six alternatives to the same annual-maxima sample: GEV by L-moments; Gumbel EV1 by sample moments; two-parameter Lognormal in natural-log space; two-parameter Gamma from the sample mean and variance; two-parameter Weibull from the sample coefficient of variation; and Log-Pearson III in base-10 log space. LP3 is an implemented comparison, not a claim of full Bulletin 17C compliance. Quantiles are compared at Gringorten plotting positions using RMSE on the 24-hour depth scale. Sherman is excluded from this distribution ranking because it is an IDF equation, not a probability model.

Sherman representation. Regress GEV return levels into a constrained power law in return period and duration with b=0. The Sherman curve is therefore a compact representation of the GEV-based IDF surface, not an independent probability distribution.

Duration and intensity conversion. For daily input, preserve 24 hours as the observation-supported duration and derive shorter durations with the selected power exponent. For hourly input, fit each rolling 1-, 2-, 3-, 6-, 12- and 24-hour annual-maxima series separately; derive only the 5-, 10-, 15- and 30-minute depths from the 1-hour fit. Divide depth by duration in hours to obtain mm/hour.

Uncertainty. At every fitted duration, resample the eligible annual-maxima series 200 times with replacement and report the 2.5th and 97.5th percentiles. These intervals describe sampling variability conditional on the selected distribution and duration-support rule.

Reference-year calculation. Calculate $\bar{y}=\Sigma y_i/n$ from the years that survive annual completeness screening. Use $\bar{y}$ by default, the last eligible y_i for the latest-supported option, or a declared custom year. Classify custom values outside $\min(y_i)$–$\max(y_i)$ as extrapolation.

Time-varying GEV. Fit stationary and nonstationary GEV models by maximum likelihood to the same annual maxima. The nonstationary model uses $\mu(y)=\mu_0+\mu_1(y-\bar{y})$ with constant scale and shape; compare models using AICc and report conditional return levels at the calculated or selected reference year with paired-bootstrap intervals.

Covariate comparison. Match annual ENSO, temperature, moisture, urbanization or another physically justified series to the same eligible annual maxima. Fit a GEV location coefficient for each candidate, compare AICc on the common years, examine probability-integral-transform residuals, quantify the coefficient with a 95% moving-block bootstrap, and evaluate the chronologically held-out final 25% of years using mean negative log-likelihood. Correlation is not used to rank or select a model.

4.4 Interpretation rule

> **Interpretation boundary.** Only the 24-hour values are directly supported by the uploaded daily series. Durations below 24 hours use P(t)=P(24h)(t/24h)^0.33 and are derived estimates, not observed sub-daily extremes.

5.0 Results and diagnostics

5.1 How to read these results

The report separates measured input from calculated output so the reader can see what the data actually support. A fitted IDF value is always a model estimate; the key distinction is whether its storm duration is supported by the uploaded interval or derived from a coarser interval.

Table 5-1. Result status and appropriate use

RESULT TYPE	WHAT IT MEANS HERE	HOW IT MAY BE USED
SUPPORTED DURATION	The uploaded daily series supports 24-hour annual maxima.	Proceed to technical review subject to data-source bias, quality checks, model assumptions and uncertainty.
DERIVED DURATION	Values below 24 hours are calculated from the 24-hour fit with exponent 0.33.	Use only when the scaling equation is acceptable for the project and its added uncertainty is acknowledged.
RARE RETURN PERIOD	The 250- and 500-year values are tail extrapolations from the fitted distribution, not storms observed every 250 or 500 years.	Use only as a high-consequence sensitivity check with method comparison and independent review.
ONLINE OR CLIMATE STATUS	Not applicable to the uploaded gauge result.	Not applicable.

5.2 Selected IDF result

For SYNTHETIC DEMONSTRATION — not measured rainfall, the selected curve family reports rainfall intensity for 2-, 5-, 10-, 25-, 50-, 100-, 250- and 500-year return periods. The estimated 100-year, 24-hour intensity is 4.04 mm/hour. The estimated 250- and 500-year values are 4.33 and 4.54 mm/hour. These last two values are model extrapolations and not observed storm magnitudes.

Intensity–Duration–Frequency curves

5 MINUTES TO 24 HOURS · 2- TO 500-YEAR RETURN PERIODS

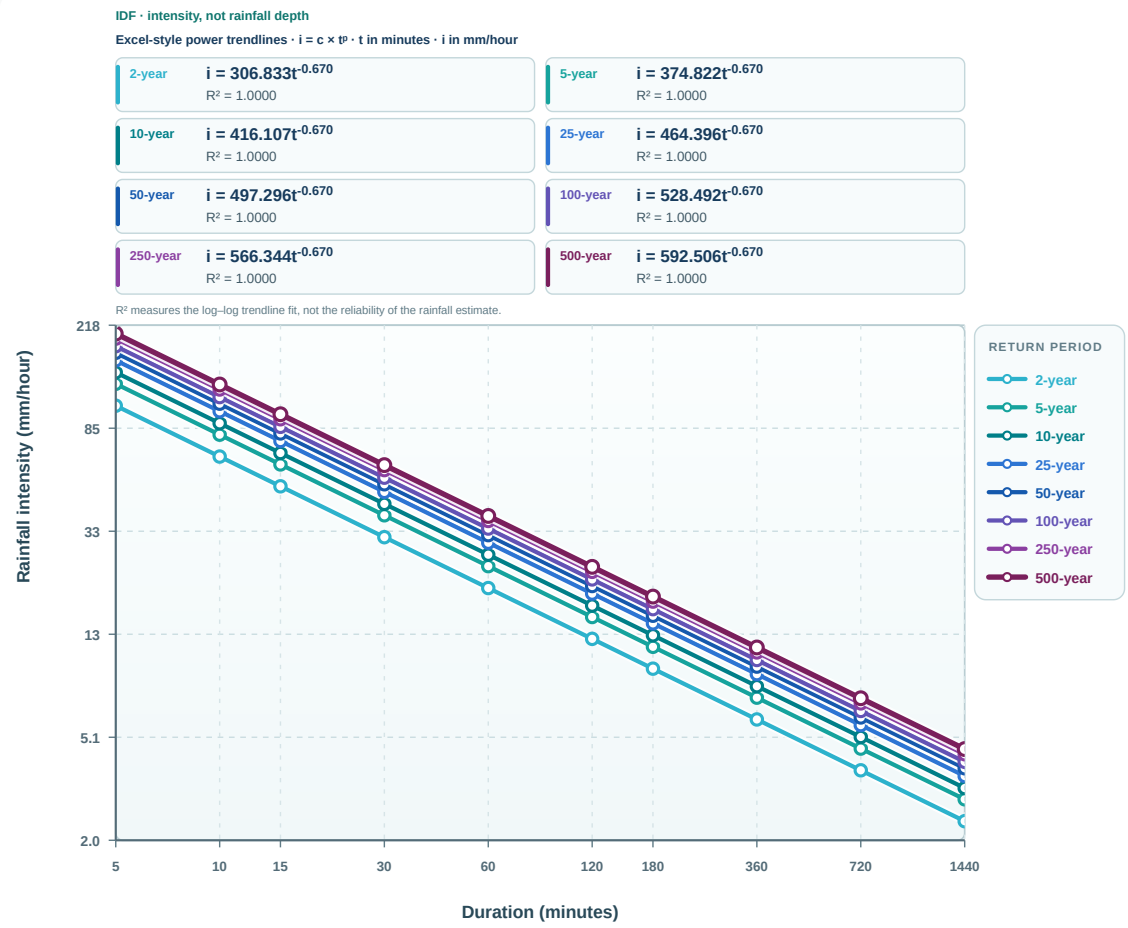


Figure 5-1. IDF curves for SYNTHETIC DEMONSTRATION — not measured rainfall, based on SYNTHETIC DEMONSTRATION — not measured rainfall · observed daily and GEV · L-moments. The fitted distribution is stationary. Duration is shown in minutes and intensity in mm/hour. The chart equation is i(t,T)=307.818T^0.114/(t+0)^0.670 (log-space R²=0.999).

Table 5-2. Selected-method 24-hour intensity and uncertainty (mm/hour)

RETURN PERIOD	INTENSITY	LOWER	UPPER	INTERVAL INTERPRETATION
2-YEAR	2.35	2.19	2.55	95% nonparametric bootstrap interval
5-YEAR	2.87	2.60	3.12	95% nonparametric bootstrap interval
10-YEAR	3.18	2.86	3.51	95% nonparametric bootstrap interval
25-YEAR	3.55	3.07	3.96	95% nonparametric bootstrap interval
50-YEAR	3.81	3.19	4.34	95% nonparametric bootstrap interval
100-YEAR	4.04	3.33	4.83	95% nonparametric bootstrap interval
250-YEAR	4.33	3.41	5.59	95% nonparametric bootstrap interval

RETURN PERIOD	INTENSITY	LOWER	UPPER	INTERVAL INTERPRETATION
500-YEAR	4.54	3.42	5.99	95% nonparametric bootstrap interval

5.3 Method comparison

Table 5-3. Comparison of six fitted probability distributions and the Sherman IDF equation at 24 hours (mm/hour)

METHOD	2-YEAR	5-YEAR	10-YEAR	25-YEAR	50-YEAR	100-YEAR	250-YEAR	500-YEAR	FIT CHECK
GEV · L-MOMENTS	2.35	2.87	3.18	3.55	3.81	4.04	4.33	4.54	RMSE 1.75 mm; normalized RMSE 3%
GUMBEL EV1 · MOMENTS	2.32	2.83	3.16	3.58	3.90	4.21	4.62	4.93	RMSE 2.05 mm; normalized RMSE 3.5%
LOG-PEARSON III	2.38	2.89	3.17	3.48	3.68	3.87	4.10	4.26	RMSE 1.94 mm; normalized RMSE 3.4%
LOGNORMAL · 2-PARAMETER	2.38	2.88	3.20	3.58	3.86	4.12	4.45	4.70	RMSE 1.77 mm; normalized RMSE 3.1%
GAMMA · 2-PARAMETER	2.37	2.88	3.17	3.50	3.73	3.94	4.20	4.39	RMSE 1.82 mm; normalized RMSE 3.1%
WEIBULL · 2-PARAMETER	2.42	2.91	3.13	3.36	3.50	3.62	3.76	3.85	RMSE 3.03 mm; normalized RMSE 5.2%
SHERMAN IDF EQUATION	2.35	2.83	3.06	3.40	3.68	3.98	4.41	4.77	RMSE 9.05 mm; normalized RMSE 15.6%

5.4 Stationarity and goodness-of-fit diagnostics

Table 5-4. 24-hour annual-maxima stationarity assessment

ELIGIBLE YEARS	35	STATUS	No significant monotonic trend
MANN–KENDALL S	-107	STANDARDIZED Z	-1.505

TWO-SIDED P-VALUE	0.1322	**DECISION AT A=0.05**	Do not reject no-trend null
SEN SLOPE	-0.340 mm/year	**NORMALIZED SLOPE**	-0.586%/year
LAG-1 CORRELATION	0.139	**SERIAL ADJUSTMENT**	Not applied

Table 5-5. Distribution fit diagnostics for 24-hour annual maxima

METHOD	SAMPLE N	RMSE (MM)	NORMALIZED RMSE	PARAMETER SUMMARY	INTERPRETATION
GEV · L-MOMENTS	35	1.75	3.0%	Location, scale and shape from L-moments	Lowest RMSE; review with probability plots
GUMBEL EV1 · MOMENTS	35	2.05	3.5%	μ=51.81, β=10.69	Eligible for comparison
LOG-PEARSON III	35	1.94	3.4%	$\log_{10}$ μ=1.751, σ=0.105, G=-0.34	Eligible for comparison
LOGNORMAL · 2-PARAMETER	35	1.77	3.1%	ln μ=4.033, ln σ=0.241	Eligible for comparison
GAMMA · 2-PARAMETER	35	1.82	3.1%	shape=17.869, scale=3.245 mm	Eligible for comparison
WEIBULL · 2-PARAMETER	35	3.03	5.2%	shape=4.826, scale=63.276 mm	Eligible for comparison
SHERMAN IDF EQUATION	35	9.05	15.6%	IDF equation calibrated to GEV levels; not a probability distribution	Eligible for comparison

Table 5-6. Stationary, time and physically plausible covariate model comparison on shared annual-maxima years

MODEL	N	AICC	ΔAICC	AKAIKE WEIGHT	COEFFICIENT (95% CI)	PIT RESIDUALS	HELD-OUT MEAN NLL	SELECTION EVIDENCE
STATIONARY GEV	35	287.17	0.00	0.596	Not applicable	Adequate; KS D=0.092; lag-1=0.110	3.8106	Reference model
TIME	35	287.95	0.77	0.404	-0.32342 (-0.98839, 0.04538) mm/year	Adequate; KS D=0.091; lag-1=0.074	3.7412	coefficient interval includes zero

Covariate decision: Inconclusive — no covariate selected. AICc, residual diagnostics, coefficient uncertainty and held-out performance do not jointly identify one covariate. A covariate is selected only when it is physically justified, within ΔAICc ≤ 2 of the best model, improves held-out log score over stationarity, has adequate PIT residual diagnostics, and its 95% coefficient interval excludes zero. Correlation is not a selection criterion. Models are ranked only on 35 common years; excluded candidates and their reasons are retained in the Excel export.

6.0 Uncertainty, limitations and robustness

Daily input directly supports only the 24-hour frequency estimate. Engineering-grade sub-daily IDF requires duration-specific observed maxima.

Automated QC does not homogenize station moves, gauge changes, observation-time changes or undocumented corrections; station metadata and neighbouring-gauge comparison remain necessary.

The Mann–Kendall test is unadjusted for serial dependence and has limited power in short records. Failure to reject the no-trend null is not proof of stationarity.

Bootstrap limits quantify sampling variability only. They exclude structural model choice, duration scaling, gauge undercatch, spatial aggregation and boundary uncertainty.

Return periods substantially longer than the eligible record are extrapolations and require sensitivity testing across methods and influential events.

District-scale interpretation requires a documented point-to-polygon or grid-area aggregation rule and a verified administrative-boundary vintage.

Report status: Suitable for hydrologic review with the stated limitations; not independently certified for design adoption.

7.0 Conclusions and recommendations

7.1 Conclusion

Stationarity is not rejected and 35 eligible maxima are available, but this is not proof of stationarity and rare return levels remain model-dependent.

7.2 Recommendations

Confirm station or product metadata, precipitation units, daily accumulation time and missing-value codes before accepting the record for design.

Use observed 5-minute, hourly or other duration-specific annual maxima for sub-daily IDF. Where unavailable, label only the 24-hour result as directly supported by the daily source.

Review probability plots, residuals, extreme-event influence and bootstrap stability across all six fitted distributions; do not choose a method from RMSE alone.

Preserve the online provider, product, retrieval time, requested geometry, grid details and returned period with every result; assess satellite or reanalysis bias against quality-controlled local gauges where available.

For time nonstationarity, center time near the record midpoint, report whether the evaluation year is interpolation or extrapolation, and compare the fitted model against the stationary alternative.

For ENSO, match an annual or seasonally justified Niño 3.4/ONI series to each annual maximum, estimate the coefficient rather than assuming it, evaluate lag and season definitions, and verify spatially varying relationships for the selected Pakistan district.

Compare time, ENSO and other physically plausible covariates using AICc, residual diagnostics, parameter uncertainty and out-of-sample performance; do not select a covariate from correlation alone.

Obtain independent hydrologic and engineering review before applying any return level to drainage, flood-control or infrastructure design.

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Appendix A · Equations and parameter definitions

GEV return level

$$z_T(y)=\mu(y)+(\sigma/\xi)[\{-\ln(1-1/T)\}^{(-\xi)}-1]$$

μ is location, σ is positive scale and ξ is shape; the Gumbel limit applies as ξ approaches zero.

Fitted time-varying location

$$\mu(y)=\mu_0+\mu_1(y-\bar{y})$$

Year is centered before maximum-likelihood fitting; scale and shape remain constant to limit over-parameterization.

Model comparison

$$AICc=2k-2\ln(L)+2k(k+1)/(n-k-1)$$

Stationary and time-varying MLE fits use the same annual maxima; $|\Delta AICc|\leq 2$ is treated as inconclusive.

Gumbel EV1 return level

$$z_T=\mu-\beta\ln\{-\ln(1-1/T)\}$$

μ is location and β is scale.

Log-Pearson III

$$\log_{10} z_T=\mu_{log}+K(T,G)\sigma_{log}$$

K depends on return period T and log-space skew G.

Two-parameter Lognormal

$$z_T=\exp[\mu_{ln}+\sigma_{ln}\,\Phi^{-1}(1-1/T)]$$

μ_ln and σ_ln are the mean and sample standard deviation of natural-log annual maxima.

Two-parameter Gamma

$$z_T=\theta\,G^{-1}(1-1/T;k)$$

G^{-1} is the unit-scale Gamma quantile; $k=(\bar{x}/s)^2$ and $\theta=s^2/\bar{x}$.

Two-parameter Weibull

$$z_T=\lambda[\ln(T)]^{(1/k)}$$

Shape k is solved from the sample coefficient of variation; λ is the scale parameter.

Sherman IDF representation

$$i(t,T)=aT^m/(t+b)^n;\ b=0$$

The equation shown on the active curve is fitted in log space to the displayed return periods and durations. It is a compact chart representation; the selected probability model remains the source of the tabulated return levels.

Kirpich time of concentration

$$t_c=0.0195L^{0.77}S^{(-0.385)}$$

L is the longest flow path in metres, S is average flow-path slope in m/m and t_c is in minutes. IDF intensity is then evaluated at storm duration t=t_c.

Automatic longest-path slope

$$S_L=(z_{up}-z_{out})/L;\ S_{10–85}=(z_{85}-z_{10})/(0.75L)$$

MGHydro mode finds the hydraulically farthest connected path in its upstream river network. Upload mode follows the selected HEC-HMS/QGIS line. Elevations are sampled from Copernicus DEM GLO-90; the selected slope is passed to Kirpich.

Design rainfall at t_c

$$i_c=aT^m/(t_c+b)^n;\ P_c=i_c\,t_c/60$$

i_c is in mm/hour and P_c is the rainfall depth in millimetres over the time of concentration.

Daily-to-duration scaling

$$P_t=P_{24}(t/24)^{0.33}$$

Values for t<24 hours are derived rather than observed.

Covariate-dependent GEV location

$$\mu(x)=\mu_0+\mu_1[(x-\bar{x})/s_x]$$

μ_1 is fitted by maximum likelihood; the reported physical-unit coefficient is μ_1/s_x with a moving-block bootstrap interval.

Appendix B · Complete IDF intensity matrix

The table reports the selected GEV · L-moments curve family. Values are rainfall intensity in mm/hour.

Table B-1. Duration by return-period intensity matrix (mm/hour)

DURATION	2-YEAR	5-YEAR	10-YEAR	25-YEAR	50-YEAR	100-YEAR	250-YEAR	500-YEAR
5 MIN	104.40	127.49	141.51	157.92	169.21	179.72	192.58	201.60
10 MIN	65.62	80.13	88.94	99.25	106.35	112.95	121.04	126.71
15 MIN	50.01	61.07	67.78	75.64	81.05	86.08	92.24	96.57
30 MIN	31.43	38.38	42.60	47.54	50.94	54.10	57.98	60.69
1 HR	19.75	24.12	26.78	29.88	32.02	34.00	36.44	38.15
2 HR	12.42	15.16	16.83	18.78	20.12	21.37	22.90	23.97
3 HR	9.46	11.55	12.83	14.31	15.34	16.29	17.45	18.27
6 HR	5.95	7.26	8.06	9.00	9.64	10.24	10.97	11.48
12 HR	3.74	4.56	5.07	5.65	6.06	6.43	6.89	7.22
24 HR	2.35	2.87	3.18	3.55	3.81	4.04	4.33	4.54

Appendix C · Assumptions and reporting controls

Table C-1. Declared analysis assumptions

TOPIC	IMPLEMENTED ASSUMPTION	REQUIRED INTERPRETATION
MAXIMUM DURATION	24 hours	Daily precipitation directly supports the 24-hour estimate only.
SUB-NATIVE DURATION	Power-law depth scaling; exponent 0.33	All durations below 24 hours are derived.
STATIONARITY DIAGNOSTIC	Two-sided Mann–Kendall test; α=0.05	Diagnostic only; non-rejection is not proof of stationarity.
NONSTATIONARY TAB	Fitted GEV location trend at 2002; AICc preference Inconclusive	Maximum-likelihood fit with constant scale and shape; ≥20 eligible maxima
REFERENCE-YEAR STATUS	Within eligible AMS years: interpolation	The year is an evaluation point, not an additional observation.
UNCERTAINTY	200 nonparametric bootstrap resamples	Sampling uncertainty only; model-form, provider bias and covariate-choice uncertainty excluded.
RARE RETURN PERIODS	250 and 500 years (0.4% and 0.2% AEP)	Highly uncertain extrapolations, especially when record length is below 50 years.
DISTRICT BOUNDARY	Not used for uploaded gauge analysis	No district polygon enters the observed-data calculation.

Appendix D · Essential terms

Only the terms required to interpret the results are defined below.

Table D-1. Plain-language definitions used in this report

TERM	MEANING IN THIS REPORT
IDF	Intensity–Duration–Frequency: rainfall intensity for a stated storm duration and annual chance of exceedance.
RETURN PERIOD AND AEP	A T-year return period means a 1/T annual exceedance probability (AEP). It is an annual chance, not a promise that the event occurs once every T years.
ANNUAL-MAXIMA SERIES	The largest qualifying rainfall accumulation retained from each eligible year and used for frequency fitting.
STATIONARY MODEL	The fitted rainfall distribution is assumed not to change with year or another explanatory variable.
NONSTATIONARY MODEL	At least one fitted distribution parameter changes with a stated explanatory variable. This report uses a time-varying GEV location when that option is selected.
REFERENCE YEAR	The year at which a time-varying model is evaluated. The automatic value is the mean year of the eligible annual maxima; it is neither an extra observation nor a forecast.
DERIVED DURATION	A duration calculated from a coarser observed interval by a stated scaling equation. It was not measured directly at that duration.
95% BOOTSTRAP INTERVAL	A range from repeated resampling of eligible annual maxima. It represents sampling variation under the fitted workflow, not every source of uncertainty.